



Fig. 4: Component layout for the PCB

transient pulses generated due to sharp chopping of the primary sine wave, RC network of suitable rating is required across the high secondary voltage as shown in Fig. 2.

Construction and testing

An actual-size, single-side PCB for the pulse generator circuit is shown in Fig. 3 and its component layout in Fig. 4.

Assemble the circuit on the provided PCB to minimise assembly errors. The PCB for EHT generator section is not provided and all the components need to be wired separately and mounted on some insulator surface. Connect G1, K1, G2 and K2 outputs on the PCB to the assembled EHT generator section corresponding to Fig. 2. Supply the input AC power and tune VRI as to

get the required output.

To verify the circuit for proper functioning, connect the mains input to transformer X1. Measure the regulated +12 V and -12 V at test points TP1 and TP2, respectively, with respect to TP0.

If the supply is alright, check full rectified waveform at TP3. This waveform is further converted into sawtooth wave, which can be verified at TP4.

Also verify the other test points (refer the table) with respect to TP0 using an oscilloscope.

EFY notes. 1. Take necessary precautions when mounting the EHT components on a proper insulator.

2. EHT transformer as well as filter chokes need at least 10kV isolation between their windings and mounting fixtures/chassis. ●

The author was director CEERI Pilani in 1989. At present he is working on developing electronics circuits for industrial applications



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